Charlie Ren

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EDUCATION

Ph.D. | Institute of Automation, Chinese Academy of Sciences (CASIA) *Sep 2025-present*
Major in Control Theory and Control Engineering
Core Modules: *Robotics, Intelligent Control, Intelligent Sensing and Information Processing, Machine Learning, Fundamentals of NLP and Large Language Models.*

Undergraduate | Ocean University of China *Sep 2021-Jun 2025*
Major in Electronic Information Engineering Rank:1/113 GPA: 3.78/4

INTERNSHIP EXPERIENCE

Zhipu AI (Beijing Zhipu Huazhang Technology Co., Ltd.) *Dec 2024-Mar 2025*
Robotics Research and Development Intern, X-Lab

- **Robotic Arm Teleoperation & Precision Grasping:** Developed a master-slave teleoperation system. Designed motion mapping algorithms from a custom controller (master) to the robotic arm (slave) and implemented Kalman filtering for jitter reduction, achieving low-latency end-effector pose tracking. Independently managed precision gripper control (error <0.5cm), reducing damage to fragile items and increasing success rates for irregular objects.
- **Tactile Sensing Data Pipeline:** Established the underlying tactile sensing architecture. Implemented high-frequency data parsing and transmission between sensors and MCUs via SPI protocol. Developed a stable, real-time signal link to support dynamic visualization and adaptive force-control grasping algorithms.
- **Automated Tray Loader Development:** Built the electrical hardware system for a tray loading machine and contributed to the motion control module. Conducted joint debugging to verify fully automated functionality.

RESEARCH EXPERIENCE

Pneumatic Flexible Robotic Arm ROV for Starfish Disaster Management *May 2022-May 2023*

- Responsible for UI interface control, pneumatic muscle production, code debugging, robot construction and testing.
- To address the issue of starfish infestation in Jiaozhou Bay, Qingdao, three types of pneumatic muscles with distinct characteristics were designed and fassbricated independently. These muscles are interconnected to form an air-pressure-powered soft robotic arm.
- The project has won one national special prize and one provincial special prize. As the first inventor, one utility model patent has been granted; two invention patents have been applied for. Meanwhile, the project was selected as a finalist of the 16th Annual Conference on Innovation and Entrepreneurship of National University Students.

High-speed Surge Defence Gun Squid Bionic Robot *May 2023-May 2024*

- As a project leader, focusing on programming, robot motion control, building communication system, and fluctuating fin fabrication.
- To solve the need for marine vehicles with low power consumption and enhanced maneuverability. Inspired by the side-fin swimming of squids, it employs sinusoidal fluctuation of specialized fins to fulfill the operational demands of underwater environment monitoring and target tracking.
- The team's achievements were interviewed and reported by CCTV, China Education Online and other mainstream media 6 times, and were exhibited on behalf of Ocean University of China in the 60th China Higher Education Expo.

PATENTS

- **Grabbing devices and fishing robots**, ZL 2023 2 1374965.8, Utility Patent, First inventor. **Granted**
- **Fishing robot**, 202310639825.7, Invention Patent, First inventor. **Granted**
- **Grabbing devices and fishing robot**, 202310640664.3, Invention Patent, Second inventor. **Granted**

REWARDS

Competition awards:

- the 12th National Marine Vehicle Design and Fabrication Competition B2-1 (Design and Fabrication of Underwater Designated Functions), National Grand Prize, 2023.

- the 12th National Marine Vehicle Design and Fabrication Competition F1(modern sailing yacht racing), National Grand Prize, 2023.
- the 12th National Marine Vehicle Design and Fabrication Competition B2-1 (Design and Fabrication of Underwater Designated Functions), National Grand Prize, 2022.
- The 3rd International Offshore Engineering Equipment Technology Innovation Competition B1 (Underwater Design Category), National Grand Prize, 2022.
- The 3rd International Offshore Engineering Equipment Technology Innovation Competition B2 (Surface Design Category), National Grand Prize, 2022.
- China International Ocean Underwater Robotics Competition Aquatic Fishing Class, National First Prize, 2023.
- the 12th National Marine Vehicle Design and Fabrication Competition F1(modern sailing yacht racing), National First Prize, 2022.
- the 11th National Marine Vehicle Design and Fabrication Competition C3-1 (Intelligent Sensing at Sea), National Second Prize, 2022.
- The 5th National University Students' Ship Energy and Power Innovation Competition A(Energy and Power Innovation), Provincial Grand Prize, 2022.
- Shandong University Physics Competition, Provincial First Prize, 2022.

Scholarships and Honors:

- National Scholarships three times, China, Dec 2022, Dec 2023 and Dec 2024.
- Wenyuan Scholarship, Ocean University of China, Nov 2024 (**Top 3** students university-wide).
- First-Class Scholarship three times, Ocean University of China, Nov 2022 and Nov 2023.
- Outstanding Students of Higher Education Institutions in Shandong Province, Shandong, May 2023.
- OUC Outstanding Student Exemplar, Ocean University of China, Nov 2024 (**Top 10** students university-wide).
- Outstanding Student, Ocean University of China, Nov 2022.
- Outstanding Student Leader, Ocean University of China, Nov 2022.
- Outstanding Youth Volunteer, Ocean University of China, Nov 2022.

WORK EXPERIENCE

Ocean University of China, Electronic Information Class 2
Class monitor and Deputy league secretary *Sep 2021-Jun 2025*

- The class was awarded the advanced class collective of Shandong University
- Class level League Branch was awarded "Red Flag League Branch" by Ocean University of China

ADDITIONAL INFORMATION

- Languages: Mandarin (Native), English (Fluent), CET4:527/CET6:487.
- Programming: C, C++, Python, Matlab, VerilogHDL.
- Embedded system development and ROS, IsaacSim, Gazebo in Linux.
- Hobbies: Basketball, Guitar.